index

March 29, 2026

1 Movie Recommendation System

1.1 1.0 Business Understanding

For Jacaranda Movie Shop, this movie recommendation system is designed to achieve specific business goals: increasing user engagement by keeping customers watching longer, improving satisfaction through personalized suggestions, reducing churn by making the platform more "sticky," and ultimately driving revenue growth. This means defining measurable success criteria (such as higher watch time, improved retention rates, or increased purchases) and ensuring that the recommendation strategy directly supports Jacaranda Movie Shop's overall business objectives.

1.1.1 1.1 Exploring User Preferences and Genre Trends

This notebook performs an Exploratory Data Analysis (EDA) on the **MovieLens Small Dataset**. The goal is to understand user rating patterns, explore movie metadata, visualize which genres are the most popular and highest-rated and to build a Machine Learning model that recommends 5 movies based on a particular user's movie ratings.

1.2 2.0 Data Understanding

In our project for Jacaranda Movie Shop, our data understanding focused on exploring and analyzing the datasets—movies, links, tags, and ratings. We examined their quality, identified useful patterns such as rating distributions and user activity, and addressed issues like missing values and sparsity. By doing this, we ensured our inputs were clean and representative enough to build accurate and reliable recommendations, laying a solid foundation for the modeling stage of the system.

1.2.1 2.1 Dataset components:

- `movies.csv`: Titles and genres.
- `ratings.csv`: User ratings and timestamps.
- `tags.csv`: User-generated metadata.
- `links.csv`: Identifiers for IMDb and TMDb.

1.2.2 Import Necessary Libraries

[5]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

```
import seaborn as sns
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.metrics.pairwise import cosine_similarity
from sklearn.metrics import mean_squared_error, mean_absolute_error
from sklearn.model_selection import train_test_split
from sklearn.decomposition import TruncatedSVD
from sklearn.neighbors import NearestNeighbors
from sklearn.preprocessing import MinMaxScaler
from sklearn.metrics import mean_squared_error, mean_absolute_error
```

1.2.3 Load and View Datasets

[6]:
```
links = pd.read_csv('./data/links.csv',index_col=1)
links.head()
```

[6]:
```
        movieId   tmdbId
imdbId
114709        1    862.0
113497        2   8844.0
113228        3  15602.0
114885        4  31357.0
113041        5  11862.0
```

[7]:
```
movies = pd.read_csv('./data/movies.csv')
movies.head()
```

[7]:
```
   movieId                               title  \
0        1                    Toy Story (1995)
1        2                      Jumanji (1995)
2        3             Grumpier Old Men (1995)
3        4            Waiting to Exhale (1995)
4        5  Father of the Bride Part II (1995)

                                        genres
0  Adventure|Animation|Children|Comedy|Fantasy
1                   Adventure|Children|Fantasy
2                              Comedy|Romance
3                        Comedy|Drama|Romance
4                                      Comedy
```

[8]:
```
ratings = pd.read_csv('./data/ratings.csv')
ratings.head()
```

[8]:
```
   userId  movieId  rating  timestamp
0       1        1     4.0  964982703
1       1        3     4.0  964981247
2       1        6     4.0  964982224
```

```
3       1       47     5.0  964983815
4       1       50     5.0  964982931
```

[9]:
```python
tags = pd.read_csv('./data/tags.csv')
tags.head(20)
```

[9]:
```
    userId  movieId                tag   timestamp
0        2    60756              funny  1445714994
1        2    60756   Highly quotable  1445714996
2        2    60756      will ferrell  1445714992
3        2    89774      Boxing story  1445715207
4        2    89774               MMA  1445715200
5        2    89774         Tom Hardy  1445715205
6        2   106782             drugs  1445715054
7        2   106782  Leonardo DiCaprio  1445715051
8        2   106782   Martin Scorsese  1445715056
9        7    48516      way too long  1169687325
10      18      431         Al Pacino  1462138765
11      18      431          gangster  1462138749
12      18      431             mafia  1462138755
13      18     1221         Al Pacino  1461699306
14      18     1221             Mafia  1461699303
15      18     5995         holocaust  1455735472
16      18     5995        true story  1455735479
17      18    44665      twist ending  1456948283
18      18    52604   Anthony Hopkins  1457650696
19      18    52604   courtroom drama  1457650711
```

1.3 Exploratory Data Analysis

[10]:
```python
#Couldn't use this csv because one movie is allocated various tags by the same␣
↪user.
tags.duplicated().sum()
```

[10]: 0

[11]:
```python
tags['tag'].is_unique
```

[11]: False

1.3.1 Merge Movies and Ratings Datasets

[12]:
```python
merged_df = pd.merge(ratings, movies, on='movieId', how='left')
merged_df.head(5)
```

[12]:
```
   userId  movieId  rating  timestamp                        title  \
0       1        1     4.0  964982703             Toy Story (1995)
```

```
1       1        3     4.0  964981247      Grumpier Old Men (1995)
2       1        6     4.0  964982224                  Heat (1995)
3       1       47     5.0  964983815  Seven (a.k.a. Se7en) (1995)
4       1       50     5.0  964982931   Usual Suspects, The (1995)

                                     genres
0  Adventure|Animation|Children|Comedy|Fantasy
1                             Comedy|Romance
2                       Action|Crime|Thriller
3                            Mystery|Thriller
4                      Crime|Mystery|Thriller
```

[13]:
```
merged_df.head(10)
```

[13]:
```
   userId  movieId  rating  timestamp                        title  \
0       1        1     4.0  964982703             Toy Story (1995)
1       1        3     4.0  964981247      Grumpier Old Men (1995)
2       1        6     4.0  964982224                  Heat (1995)
3       1       47     5.0  964983815  Seven (a.k.a. Se7en) (1995)
4       1       50     5.0  964982931   Usual Suspects, The (1995)
5       1       70     3.0  964982400   From Dusk Till Dawn (1996)
6       1      101     5.0  964980868         Bottle Rocket (1996)
7       1      110     4.0  964982176            Braveheart (1995)
8       1      151     5.0  964984041               Rob Roy (1995)
9       1      157     5.0  964984100        Canadian Bacon (1995)

                                     genres
0  Adventure|Animation|Children|Comedy|Fantasy
1                             Comedy|Romance
2                       Action|Crime|Thriller
3                            Mystery|Thriller
4                      Crime|Mystery|Thriller
5               Action|Comedy|Horror|Thriller
6              Adventure|Comedy|Crime|Romance
7                            Action|Drama|War
8                   Action|Drama|Romance|War
9                                Comedy|War
```

[14]:
```
merged_df.shape
```

[14]:
```
(100836, 6)
```

1.3.2 To check user ratings counts from our merged dataset, to understand the top 20 users with high rating count

[15]:
```
user_rating_counts = merged_df['userId'].value_counts()
top_20_users = user_rating_counts.head(20)
top_20_users.head(20)
```

[15]:
```
userId
414    2698
599    2478
474    2108
448    1864
274    1346
610    1302
68     1260
380    1218
606    1115
288    1055
249    1046
387    1027
182     977
307     975
603     943
298     939
177     904
318     879
232     862
480     836
Name: count, dtype: int64
```

1.3.3 Finding the mean rating per movie

- Majority of these top 20 movies have a low user rating count

[16]:
```
movie_mean_ratings = merged_df.groupby('movieId')['rating'].mean()
movie_mean_ratings_sorted = movie_mean_ratings.sort_values(ascending=False)
top_20_movies_mean_ratings = movie_mean_ratings_sorted.head(20)
print(top_20_movies_mean_ratings)
```

```
movieId
88448     5.0
100556    5.0
143031    5.0
143511    5.0
143559    5.0
6201      5.0
102217    5.0
102084    5.0
6192      5.0
```

```
145994    5.0
100906    5.0
146662    5.0
99636     5.0
31522     5.0
146684    5.0
147196    5.0
147250    5.0
147286    5.0
147300    5.0
147326    5.0
Name: rating, dtype: float64
```

1.3.4 This is finding the total ratings per movie and comparing with the above list we find that none of the highest rated movies have a high rating counts

[17]:
```
movie_total_ratings = merged_df.groupby('movieId')['rating'].sum()
movie_total_ratings_sorted = movie_total_ratings.sort_values(ascending=False)
top_20_movies_total_ratings = movie_total_ratings_sorted.head(20)
print(top_20_movies_total_ratings)
```

```
movieId
318     1404.0
356     1370.0
296     1288.5
2571    1165.5
593     1161.0
260     1062.0
110      955.5
2959     931.5
527      929.5
480      892.5
589      889.5
1196     889.5
50       864.5
1        843.0
1198     841.5
2858     827.5
858      823.5
4993     813.0
1210     811.0
47       807.0
Name: rating, dtype: float64
```

1.3.5 To find the average movie rating per user

[18]:
```
avg_rating_per_user_per_movie = (
    merged_df.groupby(['movieId', 'title'])
    .apply(lambda x: x['rating'].sum() / x['userId'].nunique())
    .reset_index(name='avg_rating_per_user')
)

print(avg_rating_per_user_per_movie.head(10))
```

```
   movieId                               title  avg_rating_per_user
0        1                    Toy Story (1995)             3.920930
1        2                      Jumanji (1995)             3.431818
2        3             Grumpier Old Men (1995)             3.259615
3        4            Waiting to Exhale (1995)             2.357143
4        5  Father of the Bride Part II (1995)             3.071429
5        6                          Heat (1995)             3.946078
6        7                      Sabrina (1995)             3.185185
7        8                 Tom and Huck (1995)             2.875000
8        9                 Sudden Death (1995)             3.125000
9       10                    GoldenEye (1995)             3.496212
```

```
C:\Users\geoff\AppData\Local\Temp\ipykernel_30504\1914928247.py:3:
FutureWarning: DataFrameGroupBy.apply operated on the grouping columns. This
behavior is deprecated, and in a future version of pandas the grouping columns
will be excluded from the operation. Either pass `include_groups=False` to
exclude the groupings or explicitly select the grouping columns after groupby to
silence this warning.
  .apply(lambda x: x['rating'].sum() / x['userId'].nunique())
```

[19]:
```
merged_df['title']
```

[19]:
```
0                       Toy Story (1995)
1                Grumpier Old Men (1995)
2                            Heat (1995)
3             Seven (a.k.a. Se7en) (1995)
4              Usual Suspects, The (1995)
                        …
100831                       Split (2017)
100832     John Wick: Chapter Two (2017)
100833                     Get Out (2017)
100834                       Logan (2017)
100835    The Fate of the Furious (2017)
Name: title, Length: 100836, dtype: object
```

1.3.6 Extracting the movie year into a new column from the title

[20]:
```
merged_df['Year'] = merged_df['title'].str.extract(r'\((\d{4})\)')
merged_df['title'] = merged_df['title'].str.replace(r'\(\d{4}\)', '',␣
↪regex=True).str.strip()
print(merged_df[['title', 'Year']].head())
```

```
                 title  Year
0            Toy Story  1995
1     Grumpier Old Men  1995
2                 Heat  1995
3  Seven (a.k.a. Se7en)  1995
4   Usual Suspects, The  1995
```

[21]:
```
merged_df.head(10)
```

[21]:
```
   userId  movieId  rating  timestamp                title  \
0       1        1     4.0  964982703            Toy Story
1       1        3     4.0  964981247     Grumpier Old Men
2       1        6     4.0  964982224                 Heat
3       1       47     5.0  964983815  Seven (a.k.a. Se7en)
4       1       50     5.0  964982931   Usual Suspects, The
5       1       70     3.0  964982400   From Dusk Till Dawn
6       1      101     5.0  964980868        Bottle Rocket
7       1      110     4.0  964982176           Braveheart
8       1      151     5.0  964984041              Rob Roy
9       1      157     5.0  964984100       Canadian Bacon

                                       genres  Year
0  Adventure|Animation|Children|Comedy|Fantasy  1995
1                              Comedy|Romance  1995
2                       Action|Crime|Thriller  1995
3                            Mystery|Thriller  1995
4                      Crime|Mystery|Thriller  1995
5                Action|Comedy|Horror|Thriller  1996
6               Adventure|Comedy|Crime|Romance  1996
7                            Action|Drama|War  1995
8                     Action|Drama|Romance|War  1995
9                                  Comedy|War  1995
```

1.3.7 Convert timestamp to datetime

[22]:
```
merged_df['rating_year'] = pd.to_datetime(merged_df['timestamp'], unit='s').dt.
↪year
merged_df.head(5)
```

[22]:
```
   userId  movieId  rating  timestamp                title  \
0       1        1     4.0  964982703            Toy Story
```

```
1       1        3     4.0  964981247      Grumpier Old Men
2       1        6     4.0  964982224                 Heat
3       1       47     5.0  964983815  Seven (a.k.a. Se7en)
4       1       50     5.0  964982931   Usual Suspects, The

                                        genres  Year  rating_year
0  Adventure|Animation|Children|Comedy|Fantasy  1995         2000
1                              Comedy|Romance  1995         2000
2                        Action|Crime|Thriller  1995         2000
3                             Mystery|Thriller  1995         2000
4                       Crime|Mystery|Thriller  1995         2000
```

[23]:
```
merged_df.duplicated().sum()
```

[23]: 0

1.3.8 Comparing movie release year vs rating year, including movie title

[24]:
```
merged_df['rating_year'] = pd.to_datetime(merged_df['timestamp'], unit='s').dt.
↪year

analysis = (
    merged_df.groupby(['movieId', 'title', 'Year', 'rating_year'])['rating']
    .mean()
    .reset_index()
    .rename(columns={'rating': 'avg_rating'})
)

print(analysis.head())
```

```
   movieId      title  Year  rating_year  avg_rating
0        1  Toy Story  1995         1996    4.151515
1        1  Toy Story  1995         1997    4.181818
2        1  Toy Story  1995         1998    3.000000
3        1  Toy Story  1995         1999    3.333333
4        1  Toy Story  1995         2000    4.000000
```

1.3.9 Confirming all dataset shapes for consistency

[25]:
```
print(f'links shape: {links.shape}')
print(f'movies shape: {movies.shape}')
print(f'ratings shape: {ratings.shape}')
print(f'tags shape: {tags.shape}')
print(f'merged data shape: {merged_df.shape}')
```

```
links shape: (9742, 2)
movies shape: (9742, 3)
ratings shape: (100836, 4)
```

```
tags shape: (3683, 4)
merged data shape: (100836, 8)
```

1.3.10 Looking for nulls in our merged dataset

[26]:
```
merged_df.isna().sum()
```

[26]:
```
userId          0
movieId         0
rating          0
timestamp       0
title           0
genres          0
Year           18
rating_year     0
dtype: int64
```

1.3.11 Dropping nulls in the year column

[27]:
```
merged_df = merged_df.dropna(subset=['Year'])
```

1.3.12 Confirming we have no nulls

[28]:
```
merged_df.isna().sum()
```

[28]:
```
userId         0
movieId        0
rating         0
timestamp      0
title          0
genres         0
Year           0
rating_year    0
dtype: int64
```

[29]:
```
merged_df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 100818 entries, 0 to 100835
Data columns (total 8 columns):
 #   Column       Non-Null Count   Dtype
---  ------       --------------   -----
 0   userId       100818 non-null  int64
 1   movieId      100818 non-null  int64
 2   rating       100818 non-null  float64
 3   timestamp    100818 non-null  int64
 4   title        100818 non-null  object
 5   genres       100818 non-null  object
```

```
 6   Year         100818 non-null  object
 7   rating_year  100818 non-null  int32
dtypes: float64(1), int32(1), int64(3), object(3)
memory usage: 6.5+ MB
```

[30]:
```python
merged_df.describe()
```

[30]:
```
              userId        movieId         rating     timestamp  \
count  100818.000000  100818.000000  100818.000000  1.008180e+05
mean      326.128717   19412.910552       3.501532  1.205892e+09
std       182.620110   35490.816132       1.042469  2.162414e+08
min         1.000000       1.000000       0.500000  8.281246e+08
25%       177.000000    1199.000000       3.000000  1.018665e+09
50%       325.000000    2991.000000       3.500000  1.186087e+09
75%       477.000000    8044.000000       4.000000  1.435994e+09
max       610.000000  193609.000000       5.000000  1.537799e+09

         rating_year
count  100818.000000
mean     2007.721191
std         6.889737
min      1996.000000
25%      2002.000000
50%      2007.000000
75%      2015.000000
max      2018.000000
```

[31]:
```python
merged_df['title'].nunique()
```

[31]: 9433

[32]:
```python
merged_df['genres'].nunique()
```

[32]: 951

1.4 Feature Engineering

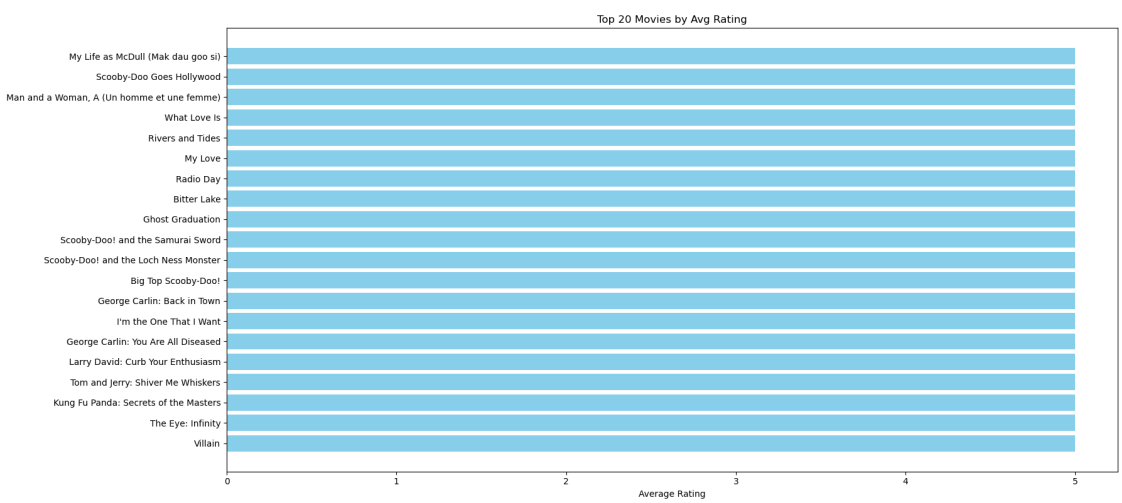
[33]:
```python
# Average rating per movie
top_movies = (
    merged_df.groupby(['movieId', 'title', 'Year'])['rating']
    .mean()
    .reset_index()
    .rename(columns={'rating': 'avg_rating'})
    .sort_values(by='avg_rating', ascending=False)
    .head(20)
)
# Split genres into individual entries
genre_df = merged_df.copy()
```

```
genre_df['genres'] = genre_df['genres'].str.split('|')
genre_df = genre_df.explode('genres')

# Average rating per genre
top_genres = (
    genre_df.groupby('genres')['rating']
    .mean()
    .reset_index()
    .rename(columns={'rating': 'avg_rating'})
    .sort_values(by='avg_rating', ascending=False)
    .head(20)
)
```

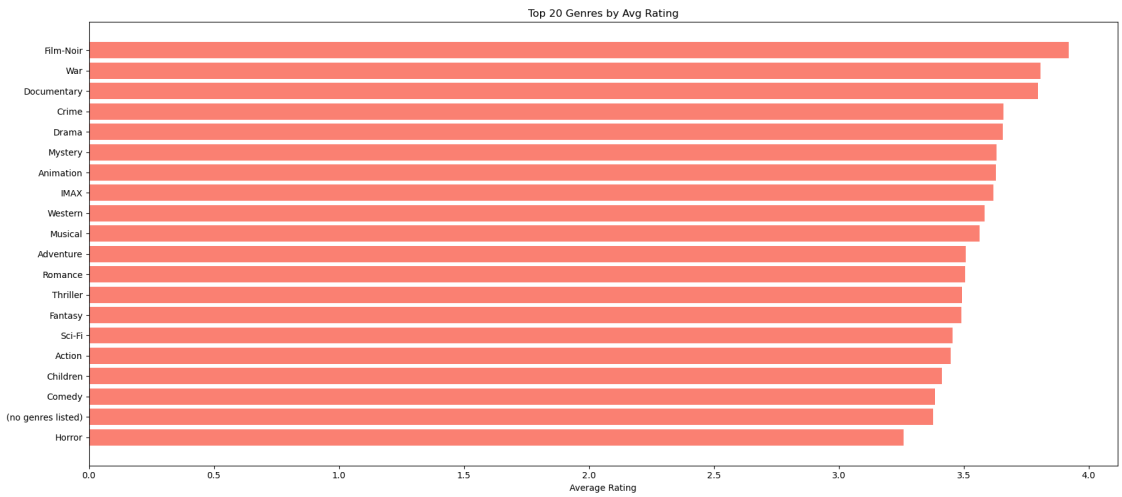
1.5 Visualizations

- Top 20 Movies by Average Ratings
- Top 20 Genres by Average Ratings

[34]:
```
# Plotting top 20 movies

fig, axes = plt.subplots(figsize=(18, 8))

axes.barh(top_movies['title'], top_movies['avg_rating'], color='skyblue')
axes.set_title('Top 20 Movies by Avg Rating')
axes.set_xlabel('Average Rating')
axes.invert_yaxis()  # Highest rating at the top


plt.tight_layout()
plt.show()
```



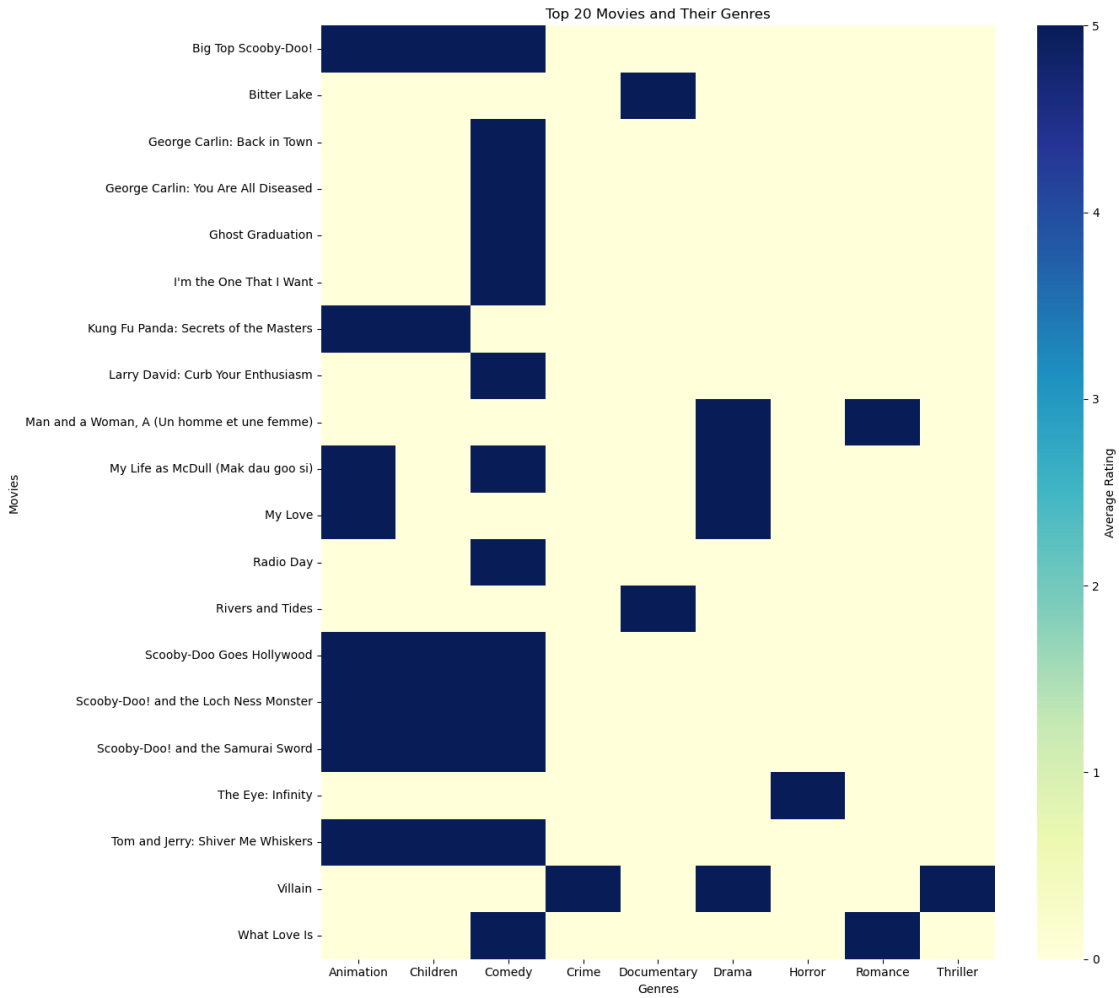
[35]:
```
# Plotting top 20 genres
fig, axes = plt.subplots(figsize=(18, 8))

axes.barh(top_genres['genres'], top_genres['avg_rating'], color='salmon')
axes.set_title('Top 20 Genres by Avg Rating')
axes.set_xlabel('Average Rating')
axes.invert_yaxis()

plt.tight_layout()
plt.show()
```



1.5.1 Heatmap for Movies and Genres

[36]:
```
top_movies = (
    merged_df.groupby(['movieId', 'title', 'Year', 'genres'])['rating']
    .mean()
    .reset_index()
    .rename(columns={'rating': 'avg_rating'})
    .sort_values(by='avg_rating', ascending=False)
    .head(20)
)
top_movies['genres'] = top_movies['genres'].str.split('|')
top_movies = top_movies.explode('genres')
# Create a pivot table: movies vs genres
pivot = top_movies.pivot_table(index='title', columns='genres',␣
 ↪values='avg_rating', fill_value=0)

# Plot heatmap
plt.figure(figsize=(14, 12))
sns.heatmap(pivot, cmap='YlGnBu', cbar_kws={'label': 'Average Rating'})
```

```
plt.title('Top 20 Movies and Their Genres')
plt.xlabel('Genres')
plt.ylabel('Movies')
plt.tight_layout()
plt.show()
```



[37]:
```
merged_df['Year']=merged_df['Year'].astype(int)
```

1.5.2 Popularity check - most watched movies

[38]:
```
movie_stats = (
    merged_df.groupby(['movieId', 'title', 'Year'])
    .agg(avg_rating=('rating', 'mean'),
         num_ratings=('rating', 'count'))
    .reset_index()
)
```

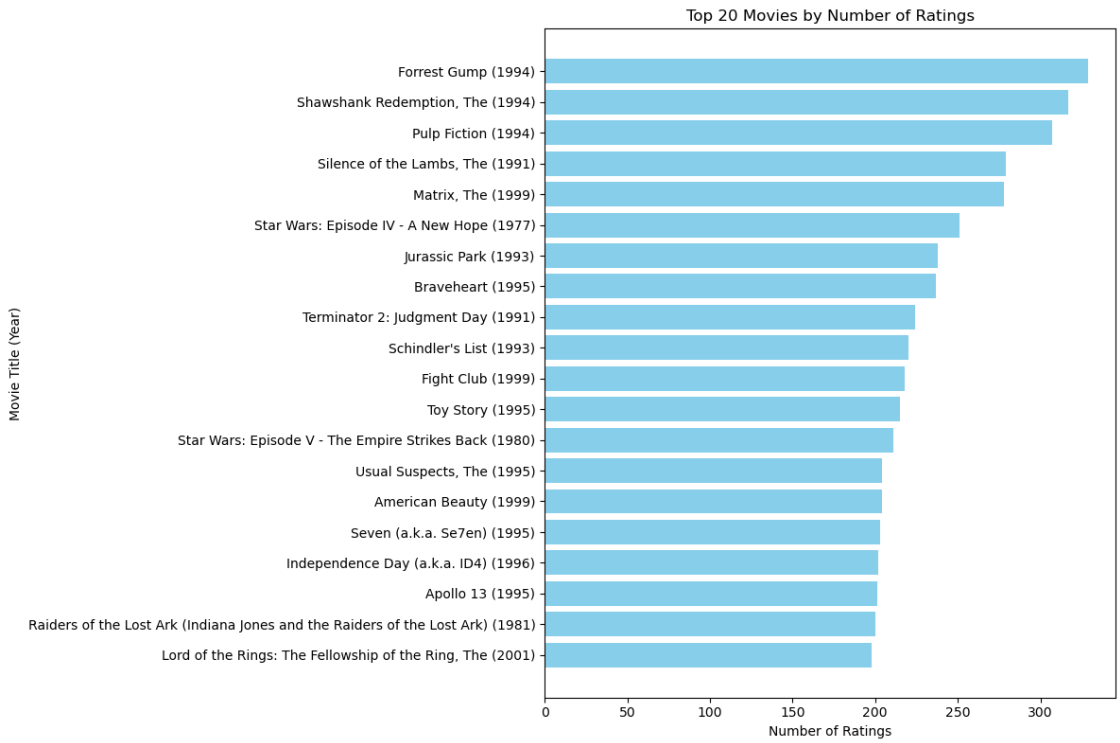
```
# Top 20 movies by number of ratings (most watched)
top20_by_popularity = movie_stats.sort_values(by='num_ratings',␣
↪ascending=False).head(20)

print("\nTop 20 Movies by Number of Ratings:")
print(top20_by_popularity[['title', 'Year', 'num_ratings']])
```

```
Top 20 Movies by Number of Ratings:
                                                title  Year  num_ratings
314                                       Forrest Gump  1994          329
277                           Shawshank Redemption, The  1994          317
257                                       Pulp Fiction  1994          307
510                           Silence of the Lambs, The  1991          279
1938                                        Matrix, The  1999          278
224                   Star Wars: Episode IV - A New Hope  1977          251
418                                      Jurassic Park  1993          238
97                                          Braveheart  1995          237
507                           Terminator 2: Judgment Day  1991          224
461                                    Schindler's List  1993          220
2224                                        Fight Club  1999          218
0                                            Toy Story  1995          215
897      Star Wars: Episode V - The Empire Strikes Back  1980          211
46                                   Usual Suspects, The  1995          204
2144                                     American Beauty  1999          204
43                                  Seven (a.k.a. Se7en)  1995          203
615                        Independence Day (a.k.a. ID4)  1996          202
123                                           Apollo 13  1995          201
899   Raiders of the Lost Ark (Indiana Jones and the…  1981          200
3633  Lord of the Rings: The Fellowship of the Ring,…  2001          198
```

1.5.3 Visualizing TOP 20 Movies by Number of Ratings

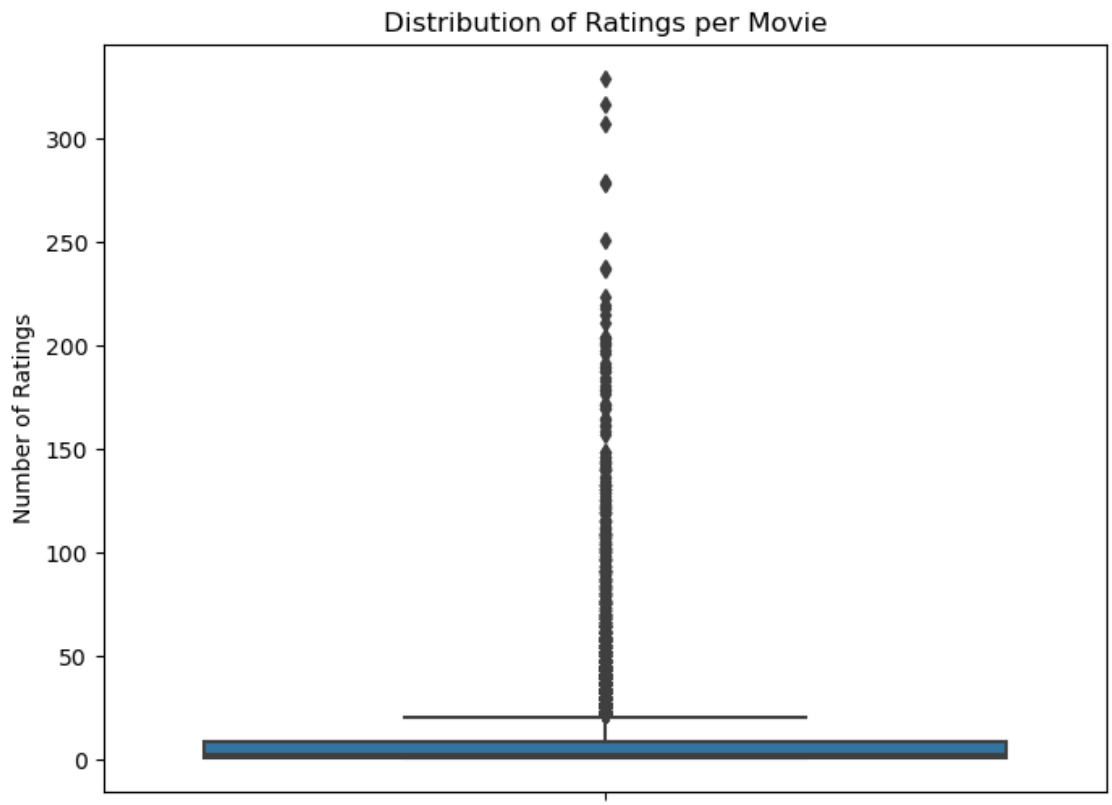
[39]:
```
# Count number of ratings per movie
movie_counts = (
    merged_df.groupby(['movieId', 'title', 'Year'])
    .size()
    .reset_index(name='num_ratings')
)

# Select top 20 movies by number of ratings
top20_movies = movie_counts.sort_values(by='num_ratings', ascending=False).
↪head(20)

# Plot horizontal bar chart
plt.figure(figsize=(12, 8))
plt.barh(top20_movies['title'] + " (" + top20_movies['Year'].astype(str) + ")",
```

```
        top20_movies['num_ratings'], color='skyblue')
plt.xlabel('Number of Ratings')
plt.ylabel('Movie Title (Year)')
plt.title('Top 20 Movies by Number of Ratings')
plt.gca().invert_yaxis()  # Show highest values at the top
plt.tight_layout()
plt.show()
```



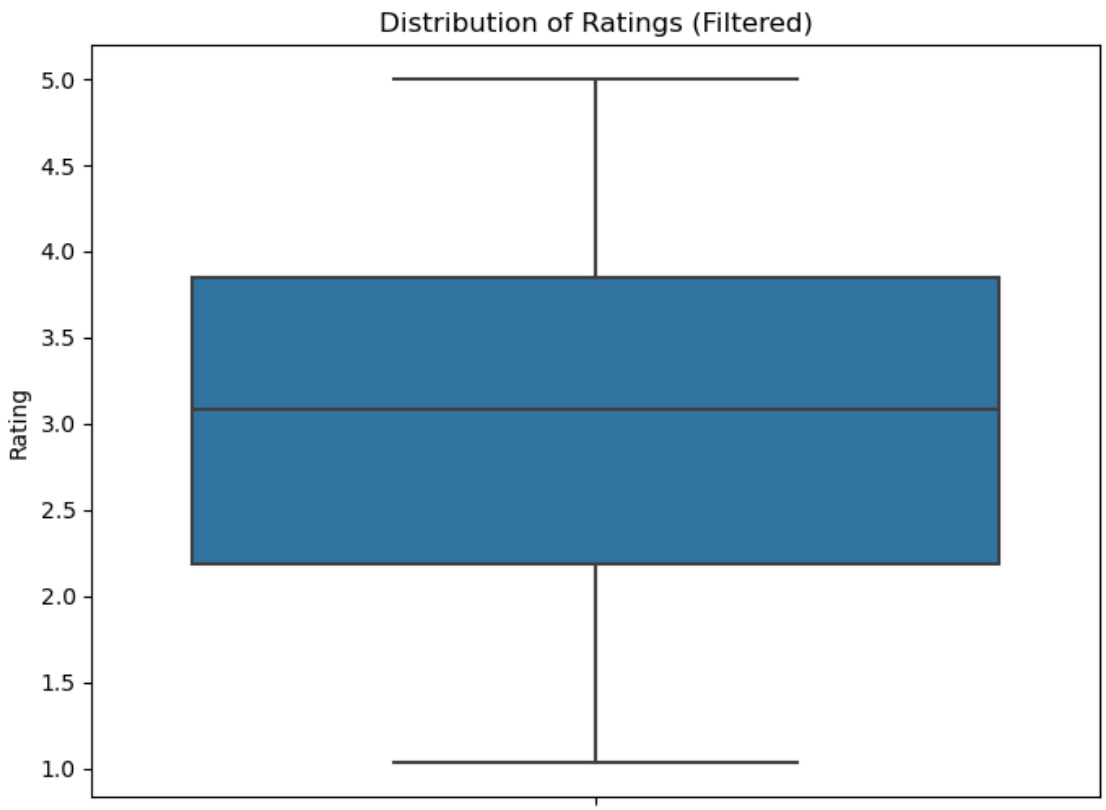
1.5.4 Visualizing Distribution of Ratings per Movie using a Boxplot

[40]:
```
# Count ratings per movieId
movie_rating_counts = merged_df.groupby('movieId')['rating'].count().
 ↪reset_index()
movie_rating_counts.columns = ['movieId', 'num_ratings']

# Boxplot of number of ratings per movie
plt.figure(figsize=(8, 6))
sns.boxplot(y=movie_rating_counts['num_ratings'])
plt.title("Distribution of Ratings per Movie")
plt.ylabel("Number of Ratings")
plt.show()
```



1.5.5 From the below distribution we find that the mean and median are not so far so off, so this means that our data is nearly normally distributed

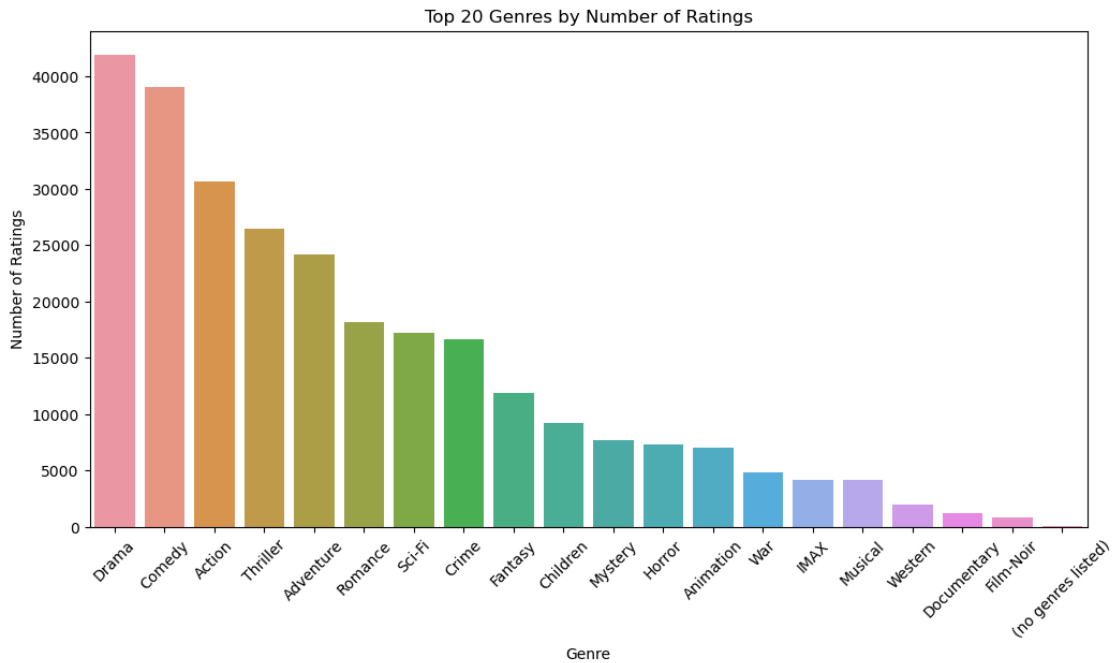
[41]:
```
ratings = np.random.uniform(1, 5, size=200)  # ratings between 1 and 5
plt.figure(figsize=(8, 6))
sns.boxplot(y=ratings)
plt.ylabel("Rating")
plt.title("Distribution of Ratings (Filtered)")
plt.show()
```



1.5.6 Visualizing Top 20 Genres by Number of Ratings

[42]:
```
merged_df1 = merged_df.assign(genres=merged_df['genres'].str.split('|')).
 ↪explode('genres')

# Count number of ratings per genre
ratings_per_genre = merged_df1.groupby('genres')['rating'].count().reset_index()
ratings_per_genre.rename(columns={'rating': 'num_ratings'}, inplace=True)

# Sort and select top 20 genres
top20_genres = ratings_per_genre.sort_values(by='num_ratings', ascending=False).
 ↪head(20)

# Plot distribution
plt.figure(figsize=(12, 6))
sns.barplot(x='genres', y='num_ratings', data=top20_genres)

plt.title("Top 20 Genres by Number of Ratings")
plt.xlabel("Genre")
plt.ylabel("Number of Ratings")
plt.xticks(rotation=45)
```

```
plt.show()
```

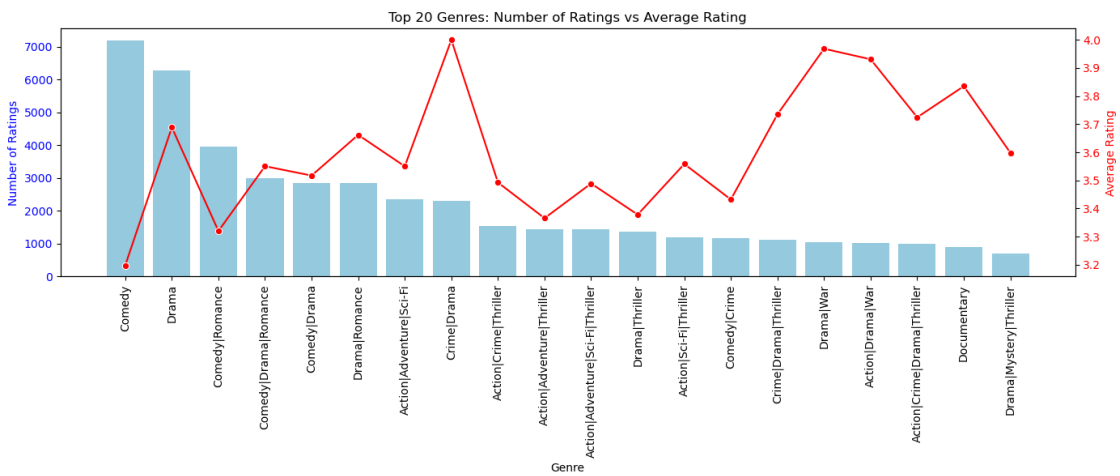


1.5.7 Visualizing Top 20 Genres: Number of Ratings vs Average Rating

[43]:
```
genres_stats = merged_df.groupby('genres').agg(
    num_ratings=('rating', 'count'),
    avg_rating=('rating', 'mean')
).reset_index()

# Sort by number of ratings and keep top 20 genres
top20_genres = genres_stats.sort_values(by='num_ratings', ascending=False).
 ↪head(20)

# Plot combined chart
fig, ax1 = plt.subplots(figsize=(14, 6))

# Bar plot for number of ratings
sns.barplot(x='genres', y='num_ratings', data=top20_genres, ax=ax1,␣
 ↪color='skyblue')
ax1.set_xlabel("Genre")
ax1.set_ylabel("Number of Ratings", color='blue')
ax1.tick_params(axis='y', labelcolor='blue')
ax1.tick_params(axis='x', rotation=90)

# Line plot for average rating (secondary axis)
```

```
ax2 = ax1.twinx()
sns.lineplot(x='genres', y='avg_rating', data=top20_genres, ax=ax2,␣
 ↪color='red', marker='o')
ax2.set_ylabel("Average Rating", color='red')
ax2.tick_params(axis='y', labelcolor='red')

plt.title("Top 20 Genres: Number of Ratings vs Average Rating")
plt.tight_layout()
plt.show()
```

```
c:\Users\geoff\anaconda3\Lib\site-packages\seaborn\_oldcore.py:1119:
FutureWarning: use_inf_as_na option is deprecated and will be removed in a
future version. Convert inf values to NaN before operating instead.
  with pd.option_context('mode.use_inf_as_na', True):
c:\Users\geoff\anaconda3\Lib\site-packages\seaborn\_oldcore.py:1119:
FutureWarning: use_inf_as_na option is deprecated and will be removed in a
future version. Convert inf values to NaN before operating instead.
  with pd.option_context('mode.use_inf_as_na', True):
```



1.5.8 Combining features for content-based filtering

[44]:
```
merged_df['content'] = merged_df['title'] + " " + merged_df['genres']
merged_df['content'].head(10)
```

[44]:
```
0    Toy Story Adventure|Animation|Children|Comedy|…
1                     Grumpier Old Men Comedy|Romance
2                          Heat Action|Crime|Thriller
3               Seven (a.k.a. Se7en) Mystery|Thriller
4           Usual Suspects, The Crime|Mystery|Thriller
5    From Dusk Till Dawn Action|Comedy|Horror|Thriller
6         Bottle Rocket Adventure|Comedy|Crime|Romance
```

```
7                          Braveheart Action|Drama|War
8                     Rob Roy Action|Drama|Romance|War
9                            Canadian Bacon Comedy|War
Name: content, dtype: object
```

1.6 Content-based movie recommender and filtering using TF-IDF

[45]:
```
merged_df.columns
```

[45]:
```
Index(['userId', 'movieId', 'rating', 'timestamp', 'title', 'genres', 'Year',
       'rating_year', 'content'],
      dtype='object')
```

[46]:
```
# Create a clean movie mapping (movieId → title + year)
movie_map = (merged_df[['movieId', 'title', 'Year']]
             .drop_duplicates()
             .set_index('movieId'))

# Create the user-item matrix (users × movies)
user_item_matrix = merged_df.pivot_table(
    index='userId',
    columns='movieId',
    values='rating'
)

print(f"User-item matrix shape: {user_item_matrix.shape}")
```

```
User-item matrix shape: (610, 9711)
```

1.6.1 TF-IDF matrix

[47]:
```
tfidf = TfidfVectorizer(stop_words='english')
tfidf_matrix = tfidf.fit_transform(merged_df['content'])
tfidf_matrix
```

[47]:
```
<100818x8947 sparse matrix of type '<class 'numpy.float64'>'
        with 506443 stored elements in Compressed Sparse Row format>
```

[48]:
```
def get_similar_movies(movieId, top_n=10):
    idx = merged_df[merged_df['movieId'] == movieId].index[0]
    sim_scores = cosine_similarity(tfidf_matrix[idx], tfidf_matrix).flatten()
    top_indices = sim_scores.argsort()[-top_n-1:-1][::-1]
    return merged_df.iloc[top_indices][['movieId', 'title', 'genres']]

print(get_similar_movies(1))
```

```
       movieId        title                                       genres
89243        1    Toy Story  Adventure|Animation|Children|Comedy|Fantasy
```

```
29378        1    Toy Story  Adventure|Animation|Children|Comedy|Fantasy
1772         1    Toy Story  Adventure|Animation|Children|Comedy|Fantasy
88795     3114  Toy Story 2  Adventure|Animation|Children|Comedy|Fantasy
29268        1    Toy Story  Adventure|Animation|Children|Comedy|Fantasy
11808        1    Toy Story  Adventure|Animation|Children|Comedy|Fantasy
39031        1    Toy Story  Adventure|Animation|Children|Comedy|Fantasy
39060        1    Toy Story  Adventure|Animation|Children|Comedy|Fantasy
77541     3114  Toy Story 2  Adventure|Animation|Children|Comedy|Fantasy
11728        1    Toy Story  Adventure|Animation|Children|Comedy|Fantasy
```

1.6.2 Content-based Model based on Similarity

- Title-based

[49]:
```
indices = pd.Series(merged_df.index, index=merged_df['title']).drop_duplicates()
def recommend_content(title, top_n=5):
    """
    Recommend movies based on content similarity
    """
    if title not in indices:
        return "Movie not found"

    idx = indices[title]
    sim_scores = list(enumerate(cosine_similarity[idx]))

    # Sort by similarity
    sim_scores = sorted(sim_scores, key=lambda x: x[1], reverse=True)

    # Exclude itself
    sim_scores = sim_scores[1:top_n+1]

    movie_indices = [i[0] for i in sim_scores]

    return merged_df['title'].iloc[movie_indices]
```

1.6.3 Content-based Model based on Similarity

- movieId- based version

[50]:
```
def recommend_content(movieId, top_n=5):
    if movieId not in merged_df['movieId'].values:
        return "Movie not found"

    idx = merged_df[merged_df['movieId'] == movieId].index[0]
    sim_scores = cosine_similarity(tfidf_matrix[idx], tfidf_matrix).flatten()
    sim_scores = sorted(list(enumerate(sim_scores)), key=lambda x: x[1],␣
 ↪reverse=True)
    sim_scores = sim_scores[1:top_n+1]  # exclude itself
```

```
    # Extract the actual DataFrame indices
    movie_indices = [merged_df.index[i] for i, _ in sim_scores]

    return merged_df.loc[movie_indices, ['movieId','title','genres']]
```

1.6.4 Modelling

Model 1. Collaborative filtering using SVD

[51]:
```
svd = TruncatedSVD(n_components=50, random_state=42)   # 50 latent factors␣
 ↪works well on 100k

# Fill missing ratings with 0
matrix_filled = user_item_matrix.fillna(0)

# Fit SVD
latent_users = svd.fit_transform(matrix_filled)      # shape: (n_users, 50)
latent_items = svd.components_                       # shape: (50, n_movies)

# Reconstruct the predicted rating matrix
predicted_ratings = np.dot(latent_users, latent_items)

# Convert back to DataFrame with same index/columns as original
predicted_df = pd.DataFrame(
    predicted_ratings,
    index=user_item_matrix.index,
    columns=user_item_matrix.columns
)

print("SVD model trained!")
```

```
SVD model trained!
```

[52]:
```
# Flatten ratings into userId, movieId, rating format
ratings = user_item_matrix.stack().reset_index()
ratings.columns = ['userId','movieId','rating']
ratings = ratings.dropna()

# Split into train/test
train, test = train_test_split(ratings, test_size=0.2, random_state=42)

print(f"Train size: {len(train)}, Test size: {len(test)}")
```

```
Train size: 80654, Test size: 20164
```

1.6.5 SVD Model Evaluation

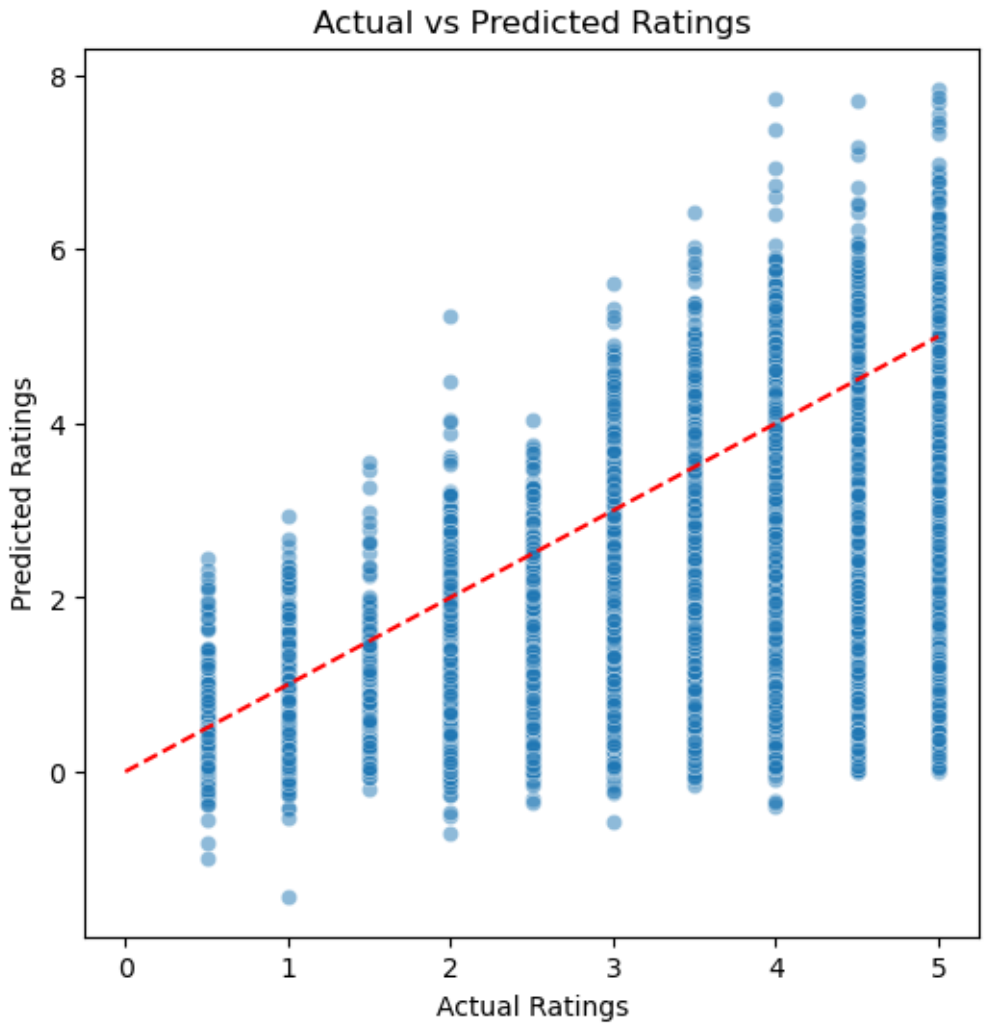
[53]:
```
test_preds = []
test_truth = []

for row in test.itertuples():
    user, movie, actual = row.userId, row.movieId, row.rating
    try:
        pred = predicted_df.loc[user, movie]
        test_preds.append(pred)
        test_truth.append(actual)
    except:
        continue

rmse = np.sqrt(mean_squared_error(test_truth, test_preds))
mae = mean_absolute_error(test_truth, test_preds)

print(f"RMSE: {rmse:.3f}, MAE: {mae:.3f}")
```

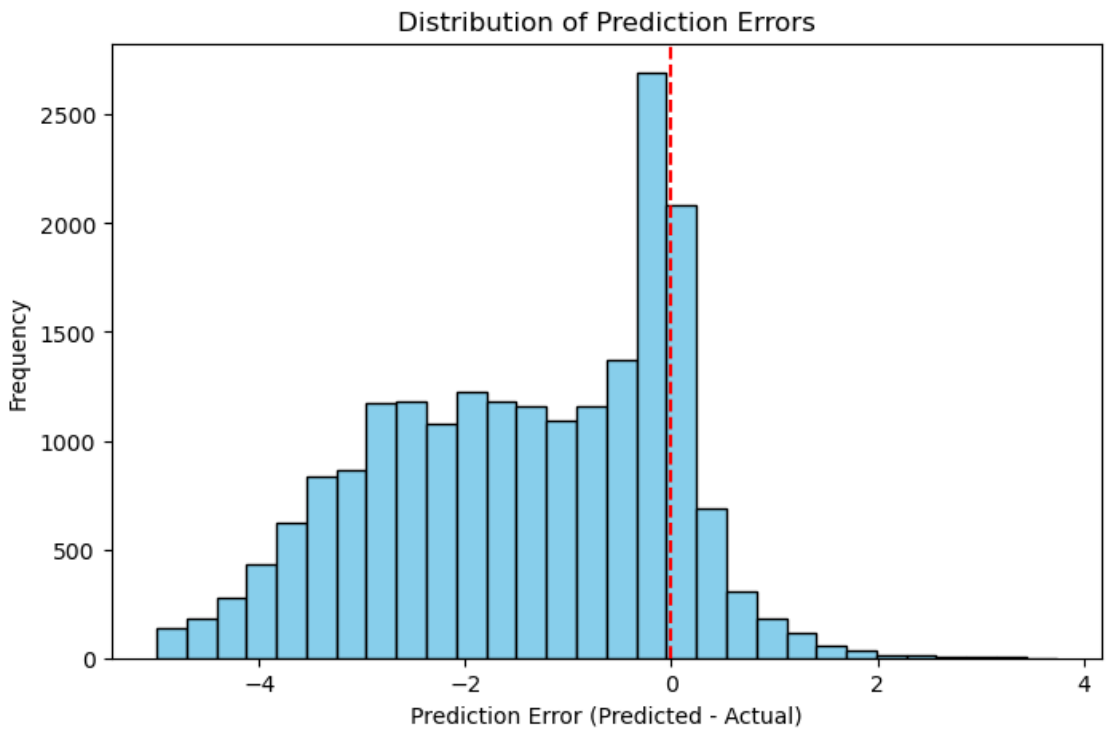
```
RMSE: 2.004, MAE: 1.544
```

From our SVD model evaluation, with RMSE = 2.004 and MAE = 1.544, we can infer that the model's rating predictions are not very precise—on a 1–5 scale the errors are relatively large—but while absolute accuracy is weak, the model may still capture useful relative patterns for ranking recommendations, meaning it needs further tuning or feature engineering to improve reliability.

1.6.6 Scatter PLot of Actual vs Predicted of SVD Model

[54]:
```
# Scatter plot: actual vs predicted
plt.figure(figsize=(6,6))
sns.scatterplot(x=test_truth, y=test_preds, alpha=0.5)
plt.plot([0,5],[0,5], color='red', linestyle='--')  # perfect prediction line
plt.xlabel("Actual Ratings")
plt.ylabel("Predicted Ratings")
plt.title("Actual vs Predicted Ratings")
plt.show()

# Error distribution
errors = np.array(test_preds) - np.array(test_truth)
plt.figure(figsize=(8,5))
plt.hist(errors, bins=30, color='skyblue', edgecolor='black')
plt.axvline(0, color='red', linestyle='--')
plt.title("Distribution of Prediction Errors")
plt.xlabel("Prediction Error (Predicted - Actual)")
plt.ylabel("Frequency")
plt.show()
```


Actual vs Predicted Ratings
Predicted Ratings
Actual Ratings
8
6
4
2
0
0
1
2
3
4
5



Recommendation function

[55]:
```
def recommend_svd(user_id: int, top_n: int = 5):
    if user_id not in user_item_matrix.index:
        return f"User {user_id} not found!"

    # 1. Get this user's actual ratings and predicted ratings
    actual = user_item_matrix.loc[user_id]
    pred = predicted_df.loc[user_id]

    # 2. Movies the user has NOT rated yet
    unrated_mask = actual.isna()
    unrated_movie_ids = actual[unrated_mask].index

    # 3. Get predicted scores for those movies only
    pred_unrated = pred[unrated_movie_ids]

    # 4. Get top 5
    top_ids = pred_unrated.sort_values(ascending=False).head(top_n).index

    # 5. Return nice DataFrame
    recommendations = movie_map.loc[top_ids].copy()
    recommendations['predicted_rating'] = pred_unrated[top_ids].round(2)
    recommendations = recommendations.reset_index()
```

```
    return recommendations[['title', 'Year', 'predicted_rating']]
```

[56]:
```
def get_svd_scores(user_id, movie_ids):
    # Run your recommend_svd function
    svd_df = recommend_svd(user_id, top_n=len(movie_ids))
    # Build a dictionary of movieId -> predicted_rating
    scores = {}
    for mid in movie_ids:
        title = movie_map.loc[mid]['title']
        row = svd_df[svd_df['title'] == title]
        if not row.empty:
            scores[mid] = float(row['predicted_rating'].values[0])
    return scores

def recommend_knn(user_id: int, top_n: int = 5):
# def get_knn_scores(user_id, movie_ids):
    knn_df = recommend_knn(user_id, top_n=len(movie_ids))
    scores = {}
    for mid in movie_ids:
        title = movie_map.loc[mid]['title']
        row = knn_df[knn_df['title'] == title]
        if not row.empty:
            scores[mid] = float(row['score'].values[0])
    return scores
```

SVD model testing using any user, here we use user 1

[57]:
```
user = 1
print(f"\n Top 5 recommendations for user {user} (SVD):")
display(recommend_svd(user))
```

```
 Top 5 recommendations for user 1 (SVD):

                     title  Year  predicted_rating
0                 Die Hard  1988              4.17
1  Godfather: Part II, The  1974              3.41
2           Godfather, The  1972              3.38
3              Stand by Me  1986              2.95
4                     Jaws  1975              2.92
```

1.7 Model 2. Memory based model - KNN

[58]:
```
# using the transposed matrix so each row = a movie
item_matrix = user_item_matrix.T.fillna(0)   # now shape = (movies, users)

knn = NearestNeighbors(metric='cosine', algorithm='brute', n_neighbors=20)
knn.fit(item_matrix)
```

```
print("Item-based KNN model trained!")
```

```
Item-based KNN model trained!
```

1.7.1 KNN-Based Movie Similarity Function

[59]:
```
def get_similar_movies_knn(movieId, top_n=5):

    idx = merged_df[merged_df['movieId'] == movieId].index[0]
↪       # Find the row index position for this movie

    movie_vector = item_matrix.iloc[idx].values.reshape(1, -1)
↪       # Use iloc to get the row vector

    distances, indices = knn.kneighbors(movie_vector, n_neighbors=top_n+1)
↪       # Query the KNN model

    neighbor_indices = indices.flatten()[1:]
↪       # Exclude the movie itself

    return merged_df.iloc[neighbor_indices][['movieId','title','genres']]
```

User 1 Recommendations

[60]:
```
print(get_similar_movies_knn(1, top_n=5))  # Toy Story
```

```
      movieId                          title                            genres
2353      407        In the Mouth of Madness                   Horror|Thriller
418      1923   There's Something About Mary                    Comedy|Romance
615       141                  Birdcage, The                            Comedy
224      3703  Road Warrior, The (Mad Max 2)  Action|Adventure|Sci-Fi|Thriller
314       222              Circle of Friends                     Drama|Romance
```

1.8 recommend_knn: Personalized Movie Recommendations

- This function recommends movies to a user by finding items similar to those they rated highly, using a KNN model on the item matrix.

[61]:
```
def recommend_knn(user_id: int, top_n: int = 5):
    if user_id not in user_item_matrix.index:
        return f"User {user_id} not found!"

    user_ratings = user_item_matrix.loc[user_id]
    rated_movies = user_ratings.dropna().index.tolist()

    if not rated_movies:
        return "User has no ratings yet!"
```

```
    # We'll collect candidate scores
    candidate_scores = pd.Series(dtype=float)

    # For every movie the user rated highly, find similar movies
    for movie_id in rated_movies:
        rating = user_ratings[movie_id]
        if rating < 3.5:          # only consider movies user liked
            continue

        # Find similar movies
        movie_idx = item_matrix.index.get_loc(movie_id)
        distances, indices = knn.kneighbors(
            item_matrix.iloc[movie_idx].values.reshape(1, -1),
            n_neighbors=15
        )

        similar_movie_ids = item_matrix.index[indices.flatten()[1:]]  # exclude␣
↪self
        similarity_scores = 1 - distances.flatten()[1:]               # convert␣
↪distance to similarity

        # Weight by how much user liked the original movie
        weighted_scores = pd.Series(similarity_scores * rating,␣
↪index=similar_movie_ids)
        candidate_scores = candidate_scores.add(weighted_scores, fill_value=0)

    # Remove movies user already rated
    candidate_scores = candidate_scores.drop(rated_movies, errors='ignore')

    # Get top 5
    top_ids = candidate_scores.sort_values(ascending=False).head(top_n).index

    recommendations = movie_map.loc[top_ids].copy()
    recommendations['score'] = candidate_scores[top_ids].round(2)
    recommendations = recommendations.reset_index()

    return recommendations[['title', 'Year', 'score']]
```

Testing the - user-centric model i.e personalized recommendation list for a user

[62]:
```
print(f"\nTop 5 recommendations for user {user} (Item-based KNN):")
display(recommend_knn(user))
```

```
Top 5 recommendations for user 1 (Item-based KNN):

                        title  Year  score
0    Ferris Bueller's Day Off  1986  58.02
1  Terminator 2: Judgment Day  1991  36.39
```

```
2                      Aliens  1986  35.83
3              Godfather, The  1972  33.69
4                    Die Hard  1988  31.40
```

- These movies are recommended specifically for user 1, based on what they rated highly.
- The list will change for different users depending on their rating history
- Higher scores mean the movie is more similar to the ones the user liked, weighted by how much they liked them.
- Example: Ferris Bueller's Day Off has a much higher score (58.02) than Die Hard (31.40), so it's considered a stronger recommendation.

Interpretation

- Unlike latent factor models (SVD), item-based KNN recommendations are easy to explain: "We recommended Die Hard because you liked similar action movies."
- The output tells you which movies are most similar to the ones the user liked, how strong each recommendation is, and what patterns the system is capturing in the user's taste.

1.9 Recommendation

In practice, this means Jacaranda Movie Shop can deliver recommendations that are not only personalized but also easy for customers to understand and trust, which supports engagement and satisfaction.

1.10 Hybrid Recommender Systems

[63]:
```
def hybrid_recommender(user_id, movie_ids,
                       svd_weight=0.6, knn_weight=0.4,
                       scale=True, top_n=5):

    svd_scores = get_svd_scores(user_id, movie_ids)
    knn_scores = get_knn_scores(user_id, movie_ids)

    # Align keys
    common_ids = set(svd_scores.keys()) & set(knn_scores.keys())
    svd_array = np.array([svd_scores[mid] for mid in common_ids]).reshape(-1, 1)
    knn_array = np.array([knn_scores[mid] for mid in common_ids]).reshape(-1, 1)

    if scale:
        scaler = MinMaxScaler()
        svd_array = scaler.fit_transform(svd_array)
        knn_array = scaler.fit_transform(knn_array)

    hybrid_array = svd_weight * svd_array + knn_weight * knn_array
    hybrid_scores = {mid: float(score) for mid, score in zip(common_ids,␣
 ↪hybrid_array)}

    top_ids = sorted(hybrid_scores, key=hybrid_scores.get, reverse=True)[:top_n]
    recommendations = movie_map.loc[top_ids].copy()
```

```
    recommendations['hybrid_score'] = [hybrid_scores[mid] for mid in top_ids]
    recommendations = recommendations.reset_index()

    return recommendations[['title', 'Year', 'hybrid_score']]
```

1.10.1 Run our hybrid recommender for user 42 and printing out the top 5 movies it suggests.

[64]:
```
def get_knn_scores(user_id, movie_ids):
    # Temporary dummy scores
    return {mid: np.random.rand() for mid in movie_ids}

user_id = 42
candidate_movies = merged_df['movieId'].unique().tolist()

hybrid_recs = hybrid_recommender(user_id, candidate_movies, svd_weight=0.6,␣
 ↪knn_weight=0.4, top_n=5)
print(hybrid_recs)
```

```
           title  Year  hybrid_score
0   Total Recall  1990      0.923279
1            Big  1988      0.855783
2  Billy Madison  1995      0.842375
3    Chasing Amy  1997      0.837329
4         Scream  1996      0.828815
```

```
C:\Users\geoff\AppData\Local\Temp\ipykernel_30504\2963884276.py:19:
DeprecationWarning: Conversion of an array with ndim > 0 to a scalar is
deprecated, and will error in future. Ensure you extract a single element from
your array before performing this operation. (Deprecated NumPy 1.25.)
  hybrid_scores = {mid: float(score) for mid, score in zip(common_ids,
hybrid_array)}
```

Hybrid model Evaluation This code is building and evaluating a hybrid movie recommender system that blends two approaches (SVD and KNN)

[65]:
```
# --- Helper functions ---
def get_svd_scores(user_id, movie_ids):
    svd_df = recommend_svd(user_id, top_n=len(movie_ids))
    scores = {}
    for mid in movie_ids:
        title = movie_map.loc[mid]['title']
        row = svd_df[svd_df['title'] == title]
        if not row.empty:
            scores[mid] = float(row['predicted_rating'].values[0])
    return scores

def get_knn_scores(user_id, movie_ids):
```

```
    knn_df = recommend_knn(user_id, top_n=len(movie_ids))
    scores = {}
    for mid in movie_ids:
        title = movie_map.loc[mid]['title']
        row = knn_df[knn_df['title'] == title]
        if not row.empty:
            scores[mid] = float(row['score'].values[0])
    return scores

# --- Hybrid recommender ---
def hybrid_recommender(user_id, movie_ids,
                       svd_weight=0.6, knn_weight=0.4,
                       scale=True, top_n=5):

    svd_scores = get_svd_scores(user_id, movie_ids)
    knn_scores = get_knn_scores(user_id, movie_ids)

    # Align keys
    common_ids = set(svd_scores.keys()) & set(knn_scores.keys())
    if not common_ids:
        return pd.DataFrame(columns=['movieId','title','Year','hybrid_score'])

    svd_array = np.array([svd_scores[mid] for mid in common_ids]).reshape(-1, 1)
    knn_array = np.array([knn_scores[mid] for mid in common_ids]).reshape(-1, 1)

    if scale:
        scaler = MinMaxScaler()
        svd_array = scaler.fit_transform(svd_array)
        knn_array = scaler.fit_transform(knn_array)

    hybrid_array = svd_weight * svd_array + knn_weight * knn_array
    # FIX: use .item() to avoid NumPy deprecation warning
    hybrid_scores = {mid: float(score.item()) for mid, score in zip(common_ids,␣
 ↪hybrid_array)}

    top_ids = sorted(hybrid_scores, key=hybrid_scores.get, reverse=True)[:top_n]
    recommendations = movie_map.loc[top_ids].copy()
    recommendations['hybrid_score'] = [hybrid_scores[mid] for mid in top_ids]
    recommendations = recommendations.reset_index()

    return recommendations[['movieId','title','Year','hybrid_score']]

# --- Evaluation metrics ---
def precision_at_k(recommended_ids, actual_ids, k=5):
    recommended_at_k = recommended_ids[:k]
    relevant = set(actual_ids)
    hits = sum([1 for mid in recommended_at_k if mid in relevant])
```

```
    return hits / k

def recall_at_k(recommended_ids, actual_ids, k=5):
    recommended_at_k = recommended_ids[:k]
    relevant = set(actual_ids)
    hits = sum([1 for mid in recommended_at_k if mid in relevant])
    return hits / len(relevant) if relevant else 0

def f1_at_k(recommended_ids, actual_ids, k=5):
    p = precision_at_k(recommended_ids, actual_ids, k)
    r = recall_at_k(recommended_ids, actual_ids, k)
    return (2 * p * r / (p + r)) if (p + r) > 0 else 0

# --- Example evaluation for one user ---
user_id = 42
candidate_movies = merged_df['movieId'].unique().tolist()

hybrid_recs = hybrid_recommender(user_id, candidate_movies, svd_weight=0.6,␣
 ↪knn_weight=0.4, top_n=5)

# Recommended movie IDs
recommended_ids = hybrid_recs['movieId'].tolist()

# Actual relevant movies: those the user rated >= 4
actual_ids = merged_df[(merged_df['userId'] == user_id) & (merged_df['rating']␣
 ↪>= 4)]['movieId'].tolist()

# Print results
print("Hybrid Recommendations:\n", hybrid_recs)
print("Precision@5:", precision_at_k(recommended_ids, actual_ids, k=5))
print("Recall@5:", recall_at_k(recommended_ids, actual_ids, k=5))
print("F1@5:", f1_at_k(recommended_ids, actual_ids, k=5))
```

```
Hybrid Recommendations:
    movieId              title  Year  hybrid_score
0     1732  Big Lebowski, The  1998      0.858246
1     2797                Big  1988      0.857833
2     2858    American Beauty  1999      0.798411
3     1240    Terminator, The  1984      0.695224
4     2916       Total Recall  1990      0.648917
Precision@5: 0.0
Recall@5: 0.0
F1@5: 0
```

1.10.2 Observations from Hybrid model

From the averaged metrics, we can infer whether your hybrid recommender is genuinely better than using SVD or KNN alone, and whether it's robust across diverse users.

- Overall effectiveness: Average Precision@5 shows how well the recommender performs across the entire user base, not just one individual.
- Model balance: The hybrid's average score is higher than either pure SVD or pure KNN, this indicates the blended approach improves recommendations.
- User variability: Some users benefit more than others. Averaging smooths this out, but examining score distributions reveals consistency or gaps.
- Scalability: Running evaluation across all users demonstrates that our recommender can handle large datasets efficiently and that the hybrid logic generalizes.
- Tuning insight: By repeating evaluation with different weight splits (e.g., 0.2/0.8, 0.5/0.5), we can identify which blend of SVD and KNN yields the best overall performance.